

BRENDAN LO

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EDUCATION

University of Chicago

B.S., Computer Science and Mathematics

Expected May 2030

Chicago, IL

Great Neck South High School

GPA 96/100 unweighted · SAT 1540

June 2026

Great Neck, NY

RESEARCH EXPERIENCE

Software Engineer Intern, Shasha Lab · New York University

Summer 2026

- Built an AI calorie-tracking app beating commercial-app accuracy with Bayesian MAP estimation, LiDAR depth capture, ARKit, and the Gemini API in Swift.

Data Analysis Intern, Kuhn & Liu Lab · Weill Cornell Medical College

Summer 2025

- Characterized calcium's role in cardiomyocyte proliferation via computational image and signal analysis in Python and Fiji/ImageJ; published in the *Journal of Cell Biology*.

Data Analysis Intern (Genomics), Long Lab · Weill Cornell Medical College

Summer 2024

- Applied RNA-seq analysis, PCR, Western blot, and cloning in R and Python to identify JARID2 gene targets and probe PRC2 in congenital heart disease.

ENGINEERING & LEADERSHIP

Co-Founder & Co-President, Web Development Club

Oct 2024 – June 2026

- Led full-stack teams across 70+ members in TypeScript, React, and Supabase; shipped and maintained 20+ sites for nonprofits, businesses, and school organizations.

Chief Technology Officer, Kudos Connect

Jan 2024 – Present

- Directed a 15-person team building the site and newsletter, reaching 7,000+ students.

Head Webmaster & Lead Programmer, FIRST Robotics Team 2638

Sep 2024 – June 2026

- Wrote trajectory-generation algorithms and robot control code; team advanced to the FRC World Championship (2025).

Software & App Lead, Programming Club

Jan 2024 – June 2026

- Taught 20+ students Python, Java, and applied ML; led 10 students building a Swift school app serving 1,300+ students.

PROJECTS

Sidewalk — 2nd Place, NYC Hackathon (Shopify × Base44)

2026

- Built a marketplace app helping NYC street vendors sell online and get licensed, and made the city's vending rules searchable for the first time by converting a 190-entry PDF into map data (Deno/TypeScript pipeline, graph search over NYC street records); shipped as an offline-capable React/Leaflet map in seven languages with 180+ automated tests.

SATsaurus — Co-Founder & Software Lead

2025 – 2026

- Led a team of five shipping a gamified SAT-prep web app in Next.js and Supabase, reaching 1,000+ students.

ML Drug-Discovery Pipeline — Lead Developer

2024 – 2025

- Ran Random Forest virtual screening and GROMACS molecular-dynamics simulations to identify PRC2-targeting drug candidates.
- Extended the work into three co-authored workshop posters at the International Conference on Machine Learning (ICML) 2026 on drug-binding prediction, retrovirus reactivation, and biological-forecasting scaling laws.

Force Network — Co-Founder & Software Lead

2022 – Present

- Built and operate a Linux hosting hub for 100+ game servers, delivering Docker deployment tooling and bandwidth to 800+ users.

PUBLICATIONS & PRESENTATIONS

Liu, H., Ammanamanchi, N., Mich-Basso, J., Panama, B., Li, Y., Huang, W., Almeida, D., Lewarchik, C., Lo, B., Wu, Y., Gotthardt, M., Kotlikoff, M., Baehr, W., Rasmusson, R., Salama, G., & Kuhn, B. (2026). Sequential changes in calcium transients during M phase regulate cardiomyocyte proliferation. *Journal of Cell Biology*. doi:10.1083/jcb.202505134

Cheng, B., Jin, A., Zhang, J., Pemmaraju, A., Lo, B., & Chang, J. (2026). A horizon-dependent intrinsic-dimension theory of scaling for biological forecasting. *HiLD Workshop, ICML 2026* (poster).

Cheng, B., Jin, A., Zhang, J., Lo, B., Pemmaraju, A., & Chang, J. (2026). ERVNet: a three-module framework for predicting endogenous retrovirus reactivation, gene propagation, and immunogenicity. *FM4LS Workshop, ICML 2026* (poster).

Cheng, B., Jin, A., Zhang, J., Pemmaraju, A., Chang, J., & Lo, B. (2026). PROTEUS: predicting how post-translational modifications alter drug binding affinity. *FM4LS Workshop, ICML 2026* (poster).

TECHNICAL SKILLS

Languages: Python, C++, Java, C#, TypeScript, JavaScript, Swift, SQL, R, Bash

Infrastructure: Linux, Docker, AWS, Google Cloud Platform, nginx, Git/GitHub, CI/CD (GitHub Actions), Playwright, Vercel

Web & mobile: React, Next.js, Node.js, Deno, Express, REST APIs, PostgreSQL, Supabase, MongoDB, SwiftUI, ARKit, Gemini API

ML & scientific: PyTorch, scikit-learn, NumPy, pandas, Bayesian MAP estimation, RNA-seq analysis, molecular docking, GROMACS, Schrödinger Maestro, Fiji/ImageJ

AWARDS & HONORS

Y Combinator Startup School 2026, selected attendee (\$25,000+ in API credits) · Eagle Scout · Science Olympiad National Qualifier and 2nd Place, New York State · Engineering Inspiration Award, FIRST Robotics World Championship · Honorable Mention, Long Island Science & Engineering Fair